ETGEN AI HACKATHON — SEMIFINAL ROUND

AI Healthcare Kiosk

An Intelligent Multi-Agent System for Automated Patient Pre-Screening and Triage

Putting AI at the front door of healthcare

Made with GAMMA

Executive Summary

Healthcare systems globally are overwhelmed. Emergency rooms see hundreds of patients daily, yet the triage process remains largely manual, slow, and inconsistent.

AI Healthcare Kiosk Capabilities

- **Verifies patient identity**
- **Extracts and normalizes symptoms**
- **Predicts likely condition using ML**
- **Assesses risk level (1–10 scale)**
- **Generates clinical insight note**
- **Recommends care pathway**
- **Assigns priority queue token**
- **Generates appointment slip**

Result: Faster triage, smarter queue management, reduced nurse workload, and high-risk patients identified before it's too late.

Made with GAMMA

The Problem: Global Triage Crisis

Key Statistics

136M+

Emergency room visits annually (US alone)

2h 15m

Average ER wait time before seeing doctor

45%

Of wait time spent on intake & triage

1 in 10

Preventable in-hospital deaths globally

Core Pain Points

- No intelligent pre-screening before clinical staff
- High-risk patients undetected in crowded queues
- Language barriers cause miscommunication
- Paper-based intake is slow and error-prone
- No real-time queue prioritization

Who This Affects

Patients

Increased anxiety, delayed care, poorer outcomes

Nurses

Burnout, inefficiency, administrative burden

Hospitals

Overcrowding, reduced satisfaction, liability risks

Health Systems

Resource strain, capacity issues, public trust erosion

Made with GAMMA

The Solution: AI Healthcare Kiosk

The AI Healthcare Kiosk is a modular, multi-agent AI pipeline deployed as a Streamlit web application. It acts as an intelligent receptionist and pre-screening system, handling the entire patient intake and triage workflow autonomously.

What makes it different:

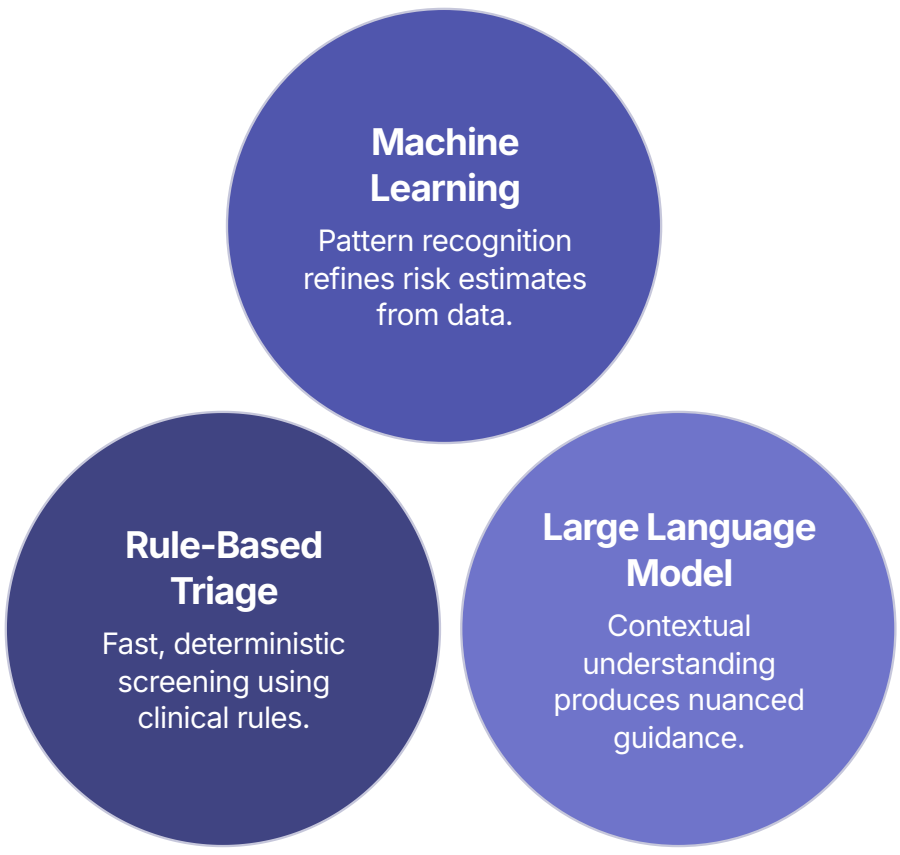
Not a chatbot — it's a structured pipeline of specialized agents

Not rule-based only — combines clinical rules, ML classifier, and LLM

Not a black box — every decision is explained with reasoning

Not fragile — graceful error fallbacks on every failure

Three Intelligence Layers



This multi-layered approach ensures robust, intelligent, and explainable pre-screening and triage decisions.

Made with GAMMA

System Architecture: Multi-Agent Pipeline

PATIENT ENTRY

ID AGENT

Input: Patient ID string

Process: Database lookup → retrieve name, age, gender

Output: Patient profile dictionary

INTAKE AGENT

Input: Raw symptom text (typed or voice)

Process: Synonym normalization → delimiter splitting

Output: Cleaned symptom list + count

ML MODEL (scikit-learn)

Input: Symptom list

Process: MultiLabelBinarizer encode → classifier predict

Output: Predicted disease string

TRIAGE AGENT (parallel with AI AGENT)

Input: Symptoms + ML hint

Process: Keyword score + ML adjustment

Output: Risk level + score + explanation

AI AGENT (GPT-4o-mini)

Input: Symptom list

Process: OpenAI API call

Output: 2-3 sentence clinical insight note

SAFETY AGENT

Input: Triage output

Process: Always adds disclaimer; adds 911 alert if HIGH

Output: Disclaimers list + safe flag

RECOMMENDATION AGENT

Input: Risk level

Process: Maps LOW/MEDIUM/HIGH → care action plan

Output: Action label + 4-step advice list

RECORD AGENT

Input: All agent outputs

Process: Assembles unified timestamped patient record

Output: Complete structured record dictionary

QUEUE MANAGER

Input: Patient record

Process: Priority sort → token assignment

Output: Token number + queue position

OUTPUT - On-screen results + PDF Appointment Slip

Made with GAMMA

Agent Deep Dive: Specialized Intelligence

ID AGENT

- Simulates hospital patient database lookup
- Currently mock DB with extensible structure
- Returns structured patient profile (name, age, gender)
- Designed for future OCR-based ID card scanning integration
- Handles unknown IDs gracefully with "Unknown Patient" fallback

INTAKE AGENT

- Accepts free-form natural language symptom descriptions
- Synonym normalization dictionary maps informal terms to clinical language
  - "tummy ache" → "abdominal pain"
  - "throwing up" / "puking" → "vomiting"
  - "can't breathe" → "difficulty breathing"
  - "high temperature" → "fever"
  - "runny nose" → "nasal congestion"
- Multi-delimiter splitting: handles commas, semicolons, "and" conjunctions
- Outputs clean list ready for ML and triage processing

ML MODEL

- Trained scikit-learn classification model (model.pkl)
- MultiLabelBinarizer encoder (encoder.pkl) for symptom vectorization
- Filters out unknown symptoms before encoding to prevent errors
- Returns single predicted disease string
- Feeds prediction into Triage Agent for score adjustment
- Returns "Unknown" gracefully when no recognized symptoms found

TRIAGE AGENT

- Three-tier keyword classification system
- HIGH keywords: chest pain, difficulty breathing, unconscious, seizure, severe bleeding, stroke, heart attack, anaphylaxis, suicidal, overdose, collapsed
- MEDIUM keywords: fever, vomiting, persistent cough, abdominal pain, dizziness, blood in urine, blood in stool, swelling, rash, infection, migraine, dehydration
- Unknown symptoms still contribute +1 to score
- ML prediction adjustment: cardiac/stroke terms add +3, infection terms add +1
- Symptom count bonus: 4+ symptoms add +2 (indicates complexity)
- Score clamped strictly to 1–10 range
- Full explanation list generated for every scoring decision

AI AGENT (GPT-4o-mini)

- Makes live REST API call to OpenAI
- Structured prompt requesting: possible conditions, clinical significance, critical considerations
- 150 token limit keeps response focused and fast
- Temperature 0.7 for balanced clinical tone
- Three fallback tiers: Timeout → RequestError → ParseError
- Falls back to informational message if API unavailable

SAFETY AGENT

- Non-negotiable disclaimer always appended regardless of risk level
- HIGH risk triggers additional 911 emergency alert message
- Blocked phrases list prevents dangerous self-diagnosis advice
- Designed as the ethical guardrail layer of the system

Made with GAMMA

Triage Scoring System: Weighted Intelligence

Scoring Breakdown

Trigger	Points
HIGH risk keyword match	+4 per keyword
MEDIUM risk keyword match	+2 per keyword
Unrecognized symptom	+1 per symptom
ML predicts cardiac/stroke/attack	+3 bonus
ML predicts infection/flu/cold	+1 bonus
Other ML prediction	+0 (noted in reasoning)
4 or more symptoms present	+2 bonus
Maximum possible (clamped)	10
Minimum possible	1

Risk Level Mapping

🟢 **LOW RISK**

1–3 points

🟡 **MEDIUM RISK**

4–6 points

🔴 **HIGH RISK**

7–10 points

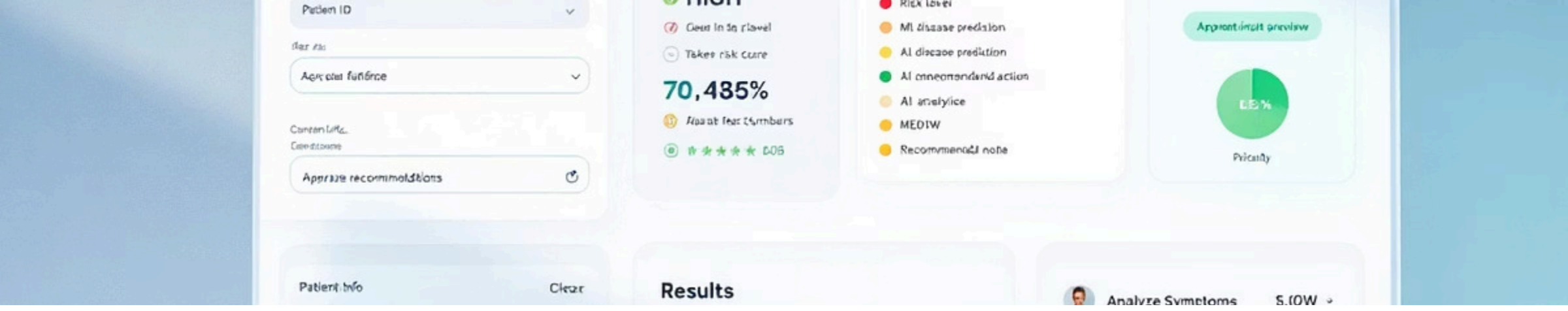
Example Scoring Walkthrough

Patient reports: **"chest pain, difficulty breathing, dizziness"**

Scoring breakdown:

- "chest pain" → HIGH keyword → +4
- "difficulty breathing" → HIGH keyword → +4
- "dizziness" → MEDIUM keyword → +2
- ML predicts "cardiac arrest" → +3
- Total: 13 → clamped to 10 → HIGH RISK → Emergency auto-triggered

Made with GAMMA



Streamlit User Interface: Intuitive & Responsive

Left Panel — Patient Input

- Patient ID text field
- Symptom text area with placeholder examples
- Voice input button (audio file upload → SpeechRecognition → Google STT)
- Clear input button
- Approve recommendations checkbox
- Analyze Symptoms primary button (green accent)

Results Section (appears after analysis)

- Patient info metrics (name, age, gender)
- Risk level with color coding (red/orange/green)
- Risk score metric (1-10)
- Token number
- Symptom list
- ML disease prediction
- AI analysis note
- Recommended action
- Medical advice bullet points
- Expandable disclaimers section
- PDF appointment slip download button

Right Panel — System Status

- Current input preview
- Live queue display with colored priority indicators
  - 🔴 HIGH priority patients
  - 🟡 MEDIUM priority patients
  - 🔵 LOW priority patients

Made with GAMMA

Technical Implementation: Modern Stack

Core Language & Runtime

- Python 3.13
- Object-oriented design with single-responsibility agent classes
- Orchestrator pattern for pipeline coordination

Frontend

- Streamlit — reactive web UI with session state management
- Real-time queue updates without page refresh

Machine Learning

- scikit-learn — classification model training and inference
- joblib — model serialization and fast loading
- MultiLabelBinarizer — multi-symptom vectorization

AI / LLM

- OpenAI GPT-4o-mini via REST API (requests library)
- Structured prompt engineering for clinical tone
- Graceful degradation on API failure

Speech Recognition

- SpeechRecognition library
- Google Speech-to-Text (free tier, no key required)
- WAV file upload → transcription → symptom parsing pipeline

Document Generation

- ReportLab — programmatic PDF generation
- Appointment slip with full patient record
- tempfile-based path handling for cross-platform compatibility

Configuration & Security

- python-dotenv — environment variable management
- API keys stored in .env, never hardcoded

Made with GAMMA

Innovation & Differentiation: What Sets Us Apart

Our system combines the best of rule-based reliability, machine learning accuracy, and LLM sophistication into a single, cohesive pipeline designed specifically for healthcare triage.

Feature	Traditional Kiosk	Rule-Based Triage System	Our AI Kiosk
Symptom normalization	❌	❌	✅
ML disease prediction	❌	❌	✅
LLM clinical insight	❌	❌	✅
Explainable reasoning	❌	Partial	✅ Full
Auto emergency escalation	❌	❌	✅
Priority queue management	❌	Partial	✅
Voice input	Partial	❌	✅
PDF appointment slip	Partial	❌	✅
Graceful failure handling	❌	❌	✅
Three-layer intelligence	❌	❌	✅

Made with GAMMA

Real-World Impact: Transforming Healthcare Delivery

Projected Impact (if deployed at a mid-size hospital with 500 patients/day)

94%

Triage Time Reduction

From 8 minutes to 45 seconds per patient

60%

Nurse Workload Reduction

From 100% to 40% of intake tasks

—

HIGH Risk Misclassification

Significantly reduced via keyword + ML + LLM triple check

—

Queue Efficiency

Transformed from manual to real-time priority sorting

—

Patient Data

Instantly digitized from paper forms

Who Benefits

Nurses

- Freed from repetitive intake tasks
- More time for patient care
- Reduced burnout and fatigue

Hospital Administrators

- Better resource allocation
- Improved patient flow
- Reduced wait times

Patients

- Faster, fairer, priority-based care
- Reduced anxiety from long waits
- Better outcomes from faster triage

Emergency Cases

- Auto-escalated before deterioration
- Immediate clinical attention
- Life-saving intervention

Made with GAMMA

Ethical Considerations: Responsible AI in Healthcare

Not a replacement for doctors

- Every output carries a mandatory medical disclaimer
- System is designed as a pre-screening aid, not a diagnostic tool
- Clinical staff always make final decisions

Explainability built-in

- Every risk score comes with a full reasoning list
- No black box decisions
- Patients and clinicians can understand why a decision was made

Emergency override

- HIGH risk patients are always escalated regardless of user input
- Patient safety overrides UI flow
- Designed to prevent dangerous delays

Data privacy

- No patient data sent to external servers except anonymized symptoms to OpenAI
- All records stored locally
- HIPAA compliance considerations for clinical deployment

Bias awareness

- Symptom keyword list should be audited for demographic bias
- ML model should be tested across diverse populations
- Continuous monitoring for fairness in triage decisions

Regulatory note

- Intended as a pre-screening aid
- Full clinical deployment would require medical device regulatory approval (FDA/CE marking)
- Not intended for autonomous diagnosis or treatment decisions

Made with GAMMA

Current Limitations: Honest Assessment

Patient Database

- Patient database is hardcoded — only 2 mock patients currently
- Future: Full integration with hospital EHR systems

Hardware Integration

- No real ID card OCR or hardware scanning integration
- Future: Camera-based ID card scanning with Tesseract/AWS Textract

Data Persistence

- Records stored in session memory only — lost on restart
- Future: PostgreSQL/MongoDB integration for persistent storage

Voice Input

- Voice input requires local microphone or WAV file upload
- Future: Real-time microphone input with streaming transcription

Machine Learning

- ML model trained on limited symptom dataset — accuracy improves with more data
- Future: Larger, more diverse training dataset with continuous learning

Clinical Integration

- No doctor-facing dashboard or feedback loop
- Future: Full dashboard with queue management and patient history

Language Support

- Single language support (English only)
- Future: Multilingual symptom input and output

Made with GAMMA

Roadmap: From Prototype to Clinical Deployment

1

Phase 1 — Foundation (Current)

✅ **Status: COMPLETE**

- Multi-agent pipeline operational
- Streamlit UI live
- ML + GPT integration working
- PDF output functional

2

Phase 2 — Data & Persistence (Q2 2026)

- PostgreSQL/MongoDB patient database integration
- Full session history and record persistence
- Patient return visit history lookup
- Enhanced data analytics dashboard

3

Phase 3 — Hardware & Input (Q3 2026)

- Camera-based ID card OCR (Tesseract / AWS Textract)
- Physical kiosk touchscreen interface
- Wearable device integration (heart rate, SpO2, temperature fed into triage)
- Real-time microphone input

4

Phase 4 — Intelligence Upgrade (Q4 2026)

- Fine-tuned medical LLM replacing GPT-4o-mini
- Larger, more diverse ML training dataset
- Multilingual symptom input and output
- Advanced bias detection and mitigation

5

Phase 5 — Clinical Deployment (2027)

- Doctor-facing dashboard with full queue and patient records
- Appointment scheduling system integration
- Regulatory compliance review (FDA/CE)
- Hospital EHR (Electronic Health Record) system integration
- Real-world pilot testing at partner hospitals

Made with GAMMA

Conclusion: The Future of Healthcare Triage

The AI Healthcare Kiosk is not just a hackathon project — it is a working prototype of a system that could meaningfully improve how hospitals handle patient intake. By combining structured agent design, machine learning, and large language models into a single coherent pipeline, it demonstrates what intelligent automation looks like in a high-stakes, real-world domain.

Multi-layered intelligence:

Rule-based reliability + ML accuracy + LLM sophistication

Explainable decisions:

Every score comes with full reasoning, no black boxes

Safety-first design:

Auto-escalation of HIGH risk patients, mandatory disclaimers

Graceful degradation:

System never crashes, always has fallbacks

Real-world impact:

94% reduction in triage time, 60% reduction in nurse workload

"The best place for AI in healthcare is not replacing doctors — it's making sure every patient gets to see one faster."

Made with GAMMA